

An applicable and efficient retrograde monosynaptic circuit mapping tool for larval zebrafish

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Abstract

The larval zebrafish is a vertebrate model for *in vivo* monitoring and manipulation of whole-brain neuronal activities. Tracing its neural circuits still remains challenging. Here we report an applicable methodology tailored for larval zebrafish to achieve efficient retrograde trans-monosynaptic tracing from genetically defined neurons via EnvA-pseudotyped glycoprotein-deleted rabies viruses. By combinatorially optimizing multiple factors involved, we identified the CVS strain trans-complemented with advanced expression of N2cG at 36°C as the optimal combination. It yielded a tracing efficiency of up to 20 inputs per starter cell. Its low cytotoxicity enabled the viable labeling and calcium imaging of infected neurons 10 days post-infection, spanning larval ages commonly used for functional examination. Cre-dependent labeling was further developed to enable input cell-type-specific tracing and circuit reconstruction. We mapped cerebellar circuits and uncovered the ipsilateral preference and subtype specificity of granule cell-to-Purkinje cell connections. Our method offers an efficient way for tracing neural circuits in larval zebrafish.

eLife assessment

This **important** study provides substantial technical development for neural circuit tracing in larval zebrafish, a widely used model for systems and developmental neurobiology, and the tool could greatly benefit neural circuit research by enabling a detailed investigation of circuit structure and function in a major model organism. The supporting evidence is **solid**, although a more detailed description of validation experiments would have increased confidence in the technique's utility. The work will interest zebrafish neurobiologists who are working on identifying novel neuronal connectivity patterns, provided that reagents generated in this study are made widely available; issues such as glial cell labeling, detailed toxicity analysis, and the impact of virus dose on tracing efficiency need further exploration to enhance the findings' applicability and robustness.

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Introduction

Given the small transparent brain and accessible genetic manipulations, the larval zebrafish has been emerging as a promising animal model for systems neuroscience. Large-scale molecular, light- and electron-microscopical (LM and EM) morphological, and physiological datasets have been gathered over the past decade to decipher the identity and connectivity of its whole-brain cells.^{1–7} Whole-brain neurons' activities can be monitored and manipulated in awake as well as behaving zebrafish larvae, providing an unprecedented research paradigm for dissecting neural mechanisms of brain functions.^{8–14} Further mechanistic insights into the roles of different cell types in brain-wide activities requires effective tools for dissecting underlying neural circuits.¹⁵

Virus-based tracing tools have opened new avenues for disentangling neural circuits.^{16,17} Among them, the EnvA-pseudotyped glycoprotein (G)-deleted rabies virus (RV) (RVdG[EnvA]) has achieved most success.^{18–20} Its success derives from being a native retrograde transsynaptic tracer engineered for cell-type-specific targeting through the recognition between EnvA and its receptor TVA²¹ and crossing only one synaptic step via trans-complementation of rabies G.^{18,19} This tool has revolutionized the structural analysis of neural circuits in rodents^{15,22} and has been continually evolved to adapt to functional studies^{23–25}. However, its application in other species including zebrafish is retarded due to lack of tailored methodologies.

The G-deleted RV (RVdG) infection of neurons in zebrafish was first demonstrated in 2009. Ten years later, RVdG[EnvA] was applied to trace input neurons in the adult zebrafish cerebellum, but the efficiency showed very low.²⁶ Recently, the herpes simplex virus (HSV1) was used as a helper virus to deliver TVA and rabies G, leading to an increased efficiency of around one input per starter neuron in the brain of adult zebrafish.²⁷ For larval zebrafish, efforts have been focused on the vesicular stomatitis virus (VSV), another rhabdovirus similar to RV but with anterograde spread.^{28–30} However, these reported VSV tools for larval zebrafish appeared relatively high cytotoxicity and have not been iterated to utilize the EnvA-TVA system to achieve initial infection specificity.^{28–30} Therefore, developing applicable efficient viral tracing methods is still an important mission in the field, and will boost the application of larval zebrafish in systems neuroscience research.

In the present study, we established applicable methodologies to implement RVdG[EnvA] for efficient retrograde trans-monosynaptic tracing from genetically defined neurons in larval zebrafish. We first demonstrated that the transient co-expression of the helper proteins TVA and rabies G in specific neurons through one-cell-stage microinjection of the GAL-UAS binary DNA plasmids could successfully help the RVdG[EnvA] microinjected later on to infect and spread from targeted starter cells. This simple two-round microinjection procedure ensures the high feasibility of our method. Then via practicing in the cerebellar circuit, we iteratively tested key factors that influence the tracing efficiency and cytotoxicity, which include the virus strain, G protein type, G expression level, and working temperature. We discovered an optimal combinatory condition: CVSdG trans-complemented with advanced expression of N2cG at 36°C. The optimality lies in the high tracing efficiency and low cytotoxicity, embodied by a 20-fold increase in efficiency compared to previous studies in zebrafish^{26,27} and a long enough time window (final fish age up to 2 - 3 weeks old) for conducting functional studies in larval zebrafish. Furthermore, by using Cre-expressing RVdG, we developed a transgenic reporter framework to enable input cell-type-specific tracing and circuit reconstruction based on single-neuron morphology. As an application demonstration, we applied the method to map the monosynaptic inputs to Purkinje cells (PCs) from granule cells (GCs) in the cerebellum and revealed the wiring relationship between morphological subtypes of GCs and PCs. These endeavors prove the applicability and efficiency of our method in dissecting interested circuits in larval zebrafish.

Results

Implementation of RVdG[EnvA]-based neural circuit tracing in larval zebrafish

To implement RVdG[EnvA]-based retrograde tracing in larval zebrafish, we first verified that larval zebrafish themselves do not express endogenous TVA. In wild type (WT) larvae without exogenous TVA expression, the injected SADdG-mCherry[EnvA], a type of RVdG[EnvA], did not infect any cells (**Figure 1—figure supplement 1A**). The observed mCherry-positive puncta co-localized with microglia (**Figure 1—figure supplement 1B**), suggesting the clearance of uninfected virus particles by microglia. These data demonstrate that, similar to mammals³¹, zebrafish do not express endogenous TVA.

Then we specifically expressed TVA in neurons (**Figures 1A** and **B**), and found that injected SADdG-mCherry[EnvA] infected TVA-expressing cells around the injection site (**Figure 1C** and **D**, dashed yellow circles). The targeted expression of TVA and the fluorescent indicator enhanced green fluorescent protein (EGFP) in neurons was achieved through transient transgene of GAL4-UAS constructs in which the GAL4 activator is driven by *elavl3*, a pan-neuronal marker in zebrafish (**Figure 1A**, and “*UGT*” in **Figure 1B**). As the G protein is necessary for viral spread, SADdG-mCherry[EnvA] did not spread beyond the initially infected neurons (i.e. starter cells) (**Figure 1C,E-G**).

We next examined whether viral particles can spread to other cells by complementing the G protein in starter neurons. As the efficiency of virus spread is reported to be positively correlated with the level of G expression in starter neurons^{32,33}, we generated two helper plasmids, *UAS:EGFP-P2A-TVA-T2A-B19G* (*UGTB*) and *UAS:EGFP-P2A-B19G-T2A-TVA* (*UGBT*), to search for a plasmid design with higher G expression while ensuring the co-expression of TVA and G (**Figure 1B**, see Methods). The position of the RV G-protein-coding gene SAD B19G was varied in the tricistronic 2A constructs for adjusting its expression level³⁴. As shown in **Figure 1D**, besides starter cells (EGFP⁺ and mCherry⁺, dashed yellow circles) and uninfected TVA-expressing cells (EGFP⁺ only), we observed mCherry⁺ only cells (arrows) of which the number increased over time, indicating the virus spread beyond the starter cells by trans-complementation with G. Notably, the

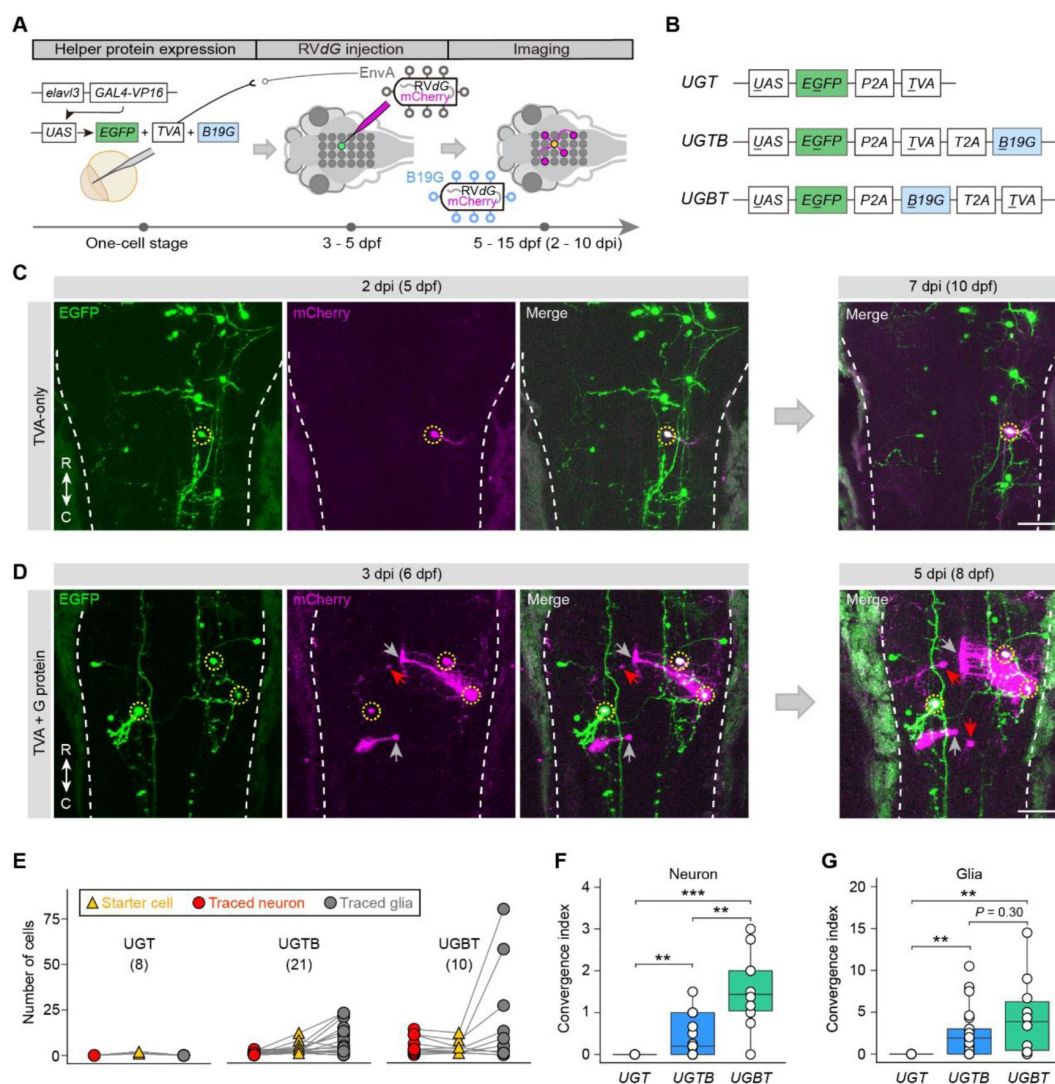


Figure 1.

Implementation of RVdG[EnvA]-based neural circuit tracing in larval zebrafish

(A) Schematic of the method used to implement RVdG[EnvA]-based circuit tracing from genetically defined cell types in larval zebrafish.

(B) Schematic of the three UAS-driven plasmids used to express the helper proteins TVA and G.

(C and D) Time-lapse confocal images of the hindbrain of larvae expressing EGFP and TVA using UGT (C) or EGFP, TVA, and B19G using UGTB (D) in randomly sparse neurons (*elavl3*⁺) and infected by SADdG-mCherry[EnvA] (magenta) injected into the hindbrain. Dashed yellow circles, starter neurons (EGFP⁺/mCherry⁺); arrows, transneuronally labeled neurons (red) and radial glia (gray) (mCherry⁺/EGFP⁻); dashed white lines, hindbrain boundaries. C, caudal; R, rostral. Scale bars, 50 μ m.

(E) Plots of the numbers of starter neurons, transneuronally traced neurons and glia in each larva expressing one of the three helper plasmids shown in (B) and injected with SADdG-mCherry[EnvA]. Data were collected between 6 and 10 dpi. Numbers in brackets indicate the number of larvae examined.

(F and G) Boxplots of the convergence index for the connection of transneuronally traced neurons (F) and glia (G) to starter neurons shown in (E). Center, median; bounds of box, first and third quartiles; whiskers, minimum and maximum values.

** $P < 0.01$, *** $P < 0.001$ (nonparametric two-tailed Mann-Whitney test).

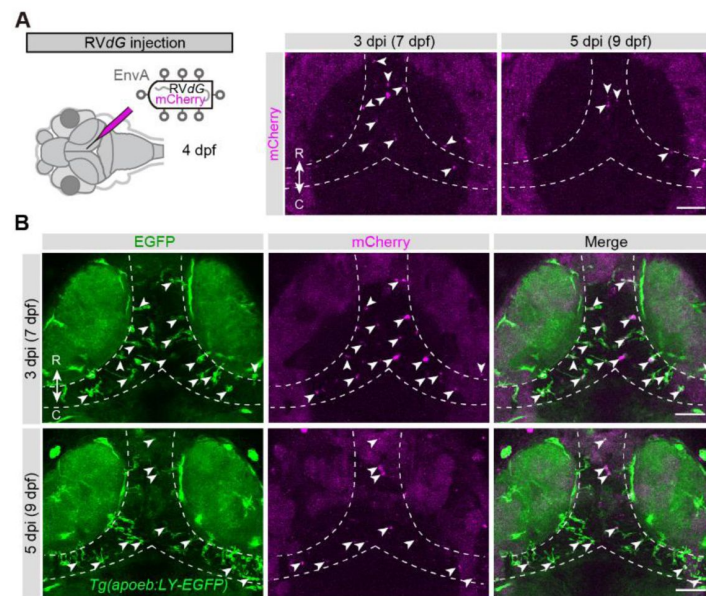


Figure 1—figure supplement 1.

No infectiousness of RVdG[EnvA] in the absence of TVA

(A) Injection of SADdG-mCherry[EnvA] into the brain of WT larvae. Left, schematic; Right, time-lapse confocal images of the optic tectum of a WT larva injected with the virus.

(B) Time-lapse confocal images of the optic tectum of a *Tg(apoeb:LY-EGFP)* larva injected with SADdG-mCherry[EnvA], showing the location of mCherry-positive puncta in EGFP-positive microglia.

Scale bars, 50 μ m. Arrowheads, fluorescent-positive punctate structure with no discernible cellular morphology under high laser excitation; dashed lines, boundaries of the cell body layer of the optic tectum. C, caudal; R, rostral.

traced cells included both neurons and radial glia (hereafter referred to as glia) identified by their distinct morphology (**Figure 1D**, red and gray arrows, respectively). Notably, the trans-infection of glia was also observed in mice^{35,36}, and its mechanism remains unknown.

Convergence index (CI), defined as the number of trans-infected cells divided by that of starter cells, was then used to estimate the tracing efficiency^{19,32}. We calculated the CI of traced neurons and glia separately, and found a higher tracing efficiency in larvae expressing *UGBT* than those expressing *UGTB* (**Figure 1E-G** and **Figure 1,2—table supplement 1**; $CI_{\text{neuron}}: 1.55 \pm 0.29$ vs 0.44 ± 0.11 , $P < 0.01$; $CI_{\text{glia}}: 4.41 \pm 1.48$ vs 2.50 ± 0.65 , $P = 0.30$; mean $\pm$ SEM), suggesting that *UGBT* is better for G expression and tracing efficiency. Taken together, these results demonstrate that in larval zebrafish, transneuronal tracing from genetically defined cell types can be implemented by combining the injection of RVdG[EnvA] with the co-expression of helper proteins mediated by the GAL4-UAS system.

Optimization and verification of the retrograde monosynaptic tracing

To optimize this system for high tracing efficiency and application feasibility in circuit research, we iterated it step-by-step in the cerebellum via testing different combinations of RV strains (SAD-B19 vs CVS-N2c), G proteins (SAD B19G vs CVS N2cG), and rearing temperatures (28°C vs 36°C) (**Figure 2A**). The cerebellar structure is evolutionarily conserved in zebrafish comparing with mammals^{37,38}. In zebrafish, Purkinje cells (PCs), the central cell type in the cerebellar circuit, receive excitatory inputs from parallel fibers and climbing fibers originated respectively from granule cells (GCs) and inferior olivary cells (IOCs), and local inhibitory inputs from stellate cells (SCs)^{37,38} (**Figure 2B**). The helper proteins G and TVA were co-expressed in PCs by using the plasmid in *UGBT* mode, driven by *cpce*, a *ca8* promoter-derived PC-specific enhancer element^{37,39} (**Figure 2A**).

Promoter	RV Strain	Helper plasmid	Glyco-protein	Temp (°C)	CI _{neuron}	CI _{glia}	CI _{all}
<i>elavl3</i> (Pan-neuron)	SAD	<i>UGT</i>	No	28	0	0	0
	SAD	<i>UGTB</i>	B19G	28	0.44 ± 0.11	2.50 ± 0.65	2.94 ± 0.72
	SAD	<i>UGBT</i>	B19G	28	1.55 ± 0.29	4.41 ± 1.48	5.97 ± 1.69
<i>cpce</i> (Purkinje cell)	SAD	<i>UGBT</i>	B19G	28	0.06 ± 0.03	0.10 ± 0.05	0.16 ± 0.06
	CVS	<i>UGBT</i>	B19G	28	0.41 ± 0.07	1.58 ± 0.23	1.99 ± 0.26
	CVS	<i>UGNT</i>	N2cG	28	1.02 ± 0.18	4.20 ± 0.94	5.22 ± 1.05
	SAD	<i>UGBT</i>	N2cG	36	4.96 ± 1.01	14.91 ± 2.49	19.87 ± 3.14
	CVS	<i>UGNT</i>	N2cG	36	11.61 ± 2.43	13.21 ± 2.44	24.82 ± 4.58
	CVS	<i>UGNT-A</i>	N2cG	36	20.35 ± 3.45	27.03 ± 4.37	47.38 ± 7.25

Figure 1,2—table supplement 1.

The efficiency of viral transfer under different tracing conditions in larval zebrafish.

The convergence index (CI) was calculated as the number of traced cells divided by the number of starter cells.

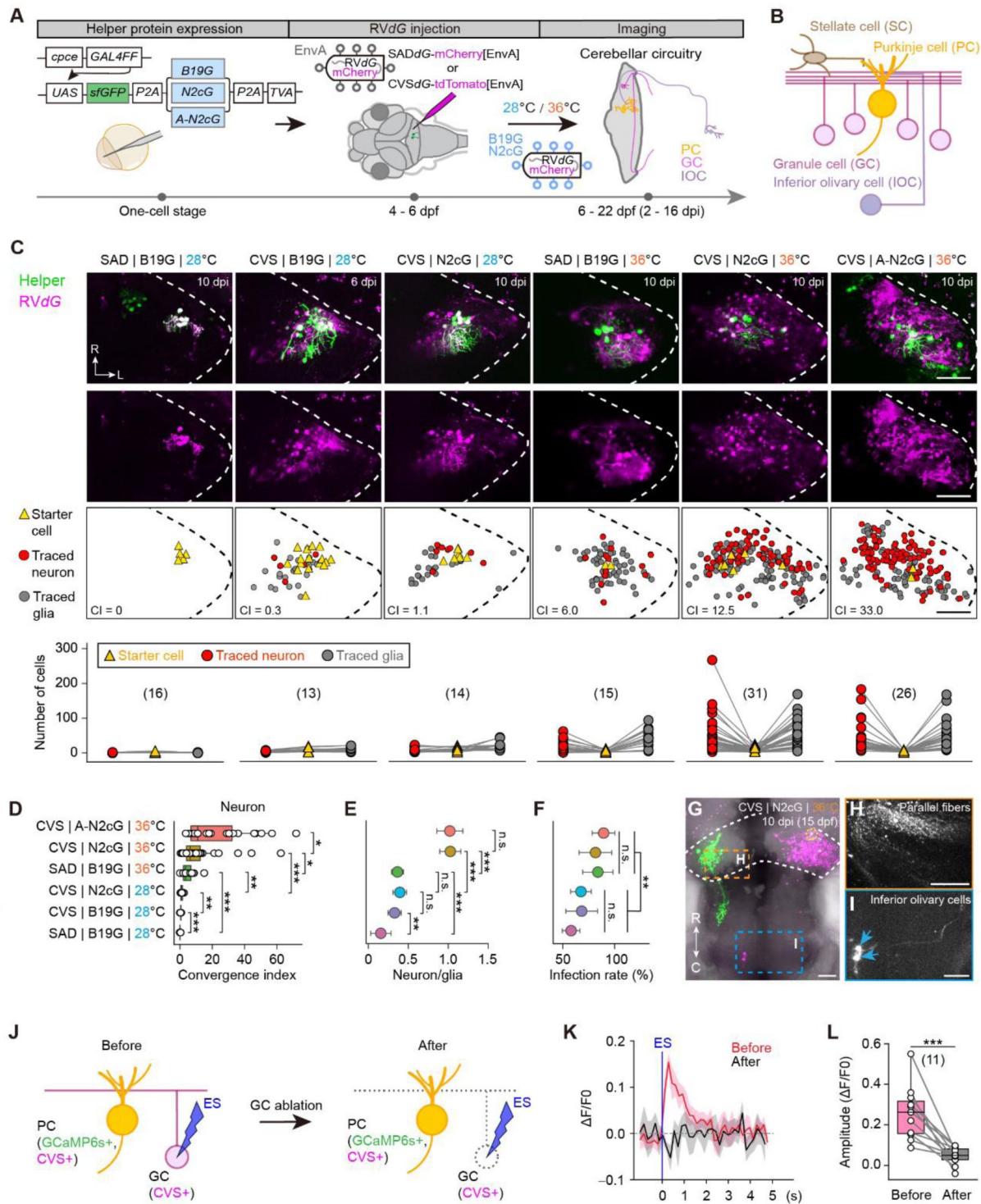


Figure 2.

Optimization and verification of RVdG[EnvA]-based retrograde monosynaptic tracing

(A) Schematic of the method used to trace inputs to PCs with RVdG[EnvA] in larval zebrafish. One of the three helper plasmids was selected to be combined with subsequent injection of SADdG-mCherry[EnvA] or CVSdG-tdTomato[EnvA]. The virus-injected larvae were raised at either 28 or 36°C, and the resulting labeling was monitored from 2 to 16 dpi.

(B) Schematic of the zebrafish cerebellar circuit related to inputs to PCs.

(C) Top, examples of *in situ* complementation in PCs under six tracing conditions. Confocal images and corresponding schematic illustrations of one cerebellar hemisphere are shown. Bottom, plots of the numbers of starter neurons, transneuronally traced neurons and glia in each larva under one of the six tracing conditions at 6 - 10 dpi. Dashed lines, the cerebellar boundaries. CI, convergence index of traced neurons. Numbers in brackets indicate the number of larvae examined.

(D-F) CI for the connection of transneuronally traced neurons to starter PCs (D), the ratio of the number of traced neurons to that of glia (E), and the proportion of infected fish (F, 3 to 5 independent experiments) under the six tracing conditions shown in (C, bottom).

(G), Confocal image of a larva infected with CVSdG-tdTomato[EnvA] that was trans-complemented with CVS N2cG at 36°C. Dashed yellow circle, starter PC; dashed white line, the cerebellar boundary.

(H and I) Enlarged view of the boxed regions in (G), showing transneuronally traced parallel fibers of GCs (H) and two IOCs (I). Blue arrows, somata of IOCs.

(J) Schematic of the experiment for verifying the functional connection between traced GCs and the starter PC in larvae infected with CVSdG-tdTomato[EnvA] that was trans-complemented with advanced CVS N2cG at 36°C. Ca^{2+} imaging was conducted on the starter PC in response to electrical stimulation (ES) at the GC position before and after two-photon laser ablation of the GC.

(K) Average Ca^{2+} response (5 trials) of the same starter PC before (red) and after (black) GC ablation. The shadowed area represents SEM.

(L) Quantification of the amplitude of PC Ca^{2+} responses before and after GC ablation. The number in bracket indicate the number of GC-PC pairs examined.

Scale bars, 50 μm . SAD/CSV|B19G/N2cG/A-N2cG|28/36°C, the tracing conditions written as the G-deleted RV strain that was injected | the helper glycoprotein used for trans-complementation | fish rearing temperature after virus injection. C, caudal; L, lateral; R, rostral. Boxplots in (D and L): center, median; bounds of box, first and third quartiles; whiskers, minimum and maximum values. n.s., not-significant; * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ (nonparametric two-tailed Mann-Whitney test in (D-F); two-tailed unpaired Student's t test in (L)). Error bars denote SEM..

With the same G protein SAD B19G and rearing temperature at 28°C, CVSdG[EnvA] exhibited around 6-fold increase in neuron tracing efficiency compared to SADdG[EnvA] (**Figure 2C,D**, **Figure 2—figure supplement 1A-C** and **Figure 1,2—table supplement 1**; $\text{CI}_{\text{neuron}}$: 0.41 ± 0.07 vs 0.06 ± 0.03 , $P < 0.001$). After replacing SAD B19G with CVS N2cG to complement CVSdG[EnvA], the tracing efficiency was further increased by ~ 2 times (**Figure 2C,D** and **Figure 1,2—table supplement 1**; $\text{CI}_{\text{neuron}}$: 1.02 ± 0.18 vs 0.41 ± 0.07 , $P < 0.01$). In addition, CVSdG[EnvA] showed a greater tendency to spread to neurons rather than glia in comparison with SADdG[EnvA] (**Figure 2C,E**). In all experiments, virus injection was performed at 4 - 6 days post-fertilization

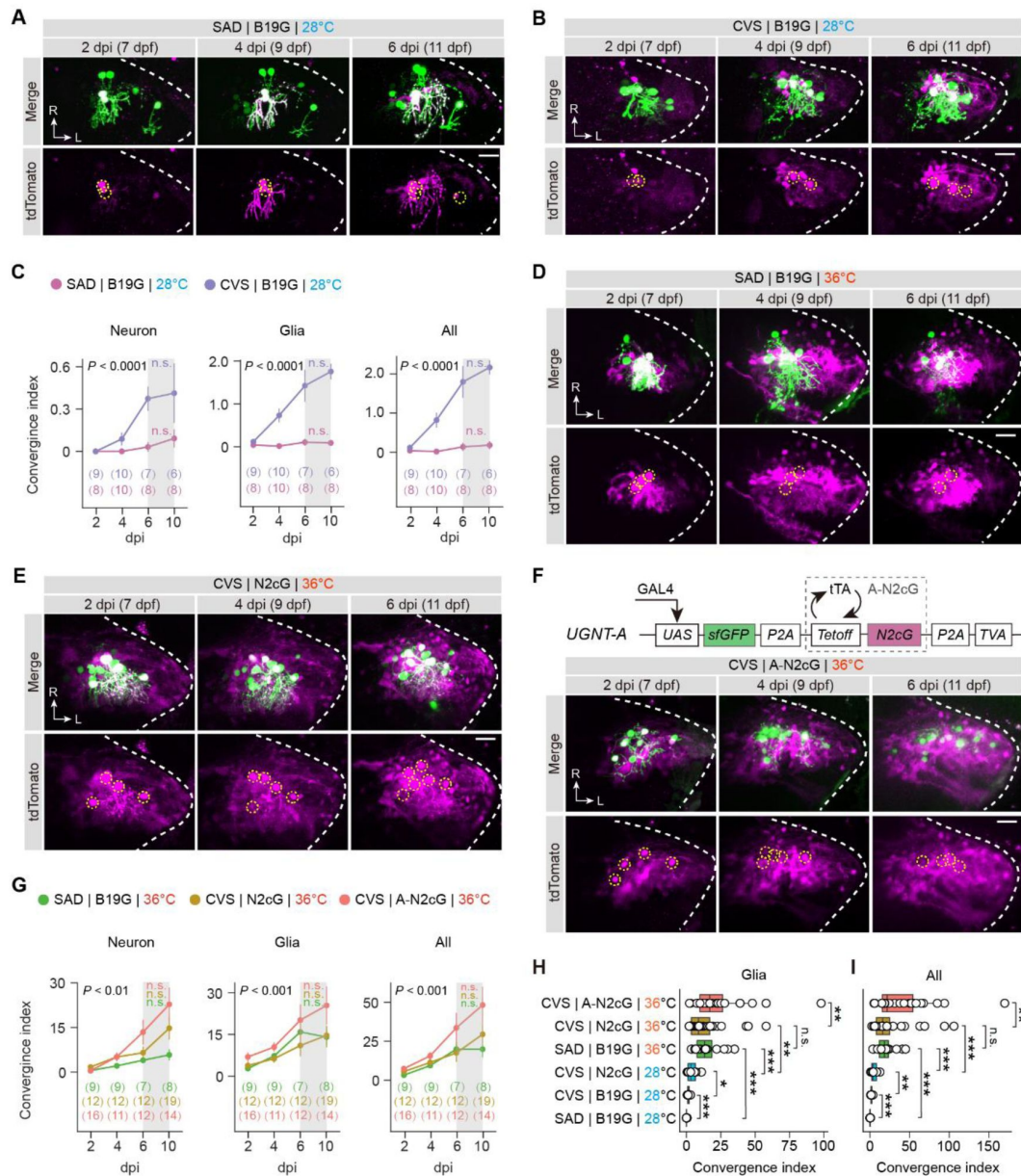


Figure 2—figure supplement 1.

Time-lapse images of trans-complemented tracing from Purkinje cells in the cerebellum under different tracing conditions

(A, B, and D-F) Time-lapse (2, 4, and 6 dpi) confocal images of *in situ* complementation in Purkinje cells (PCs) under tracing conditions of SAD|B19G|28°C (A), CVS|B19G|28°C (B), SAD|B19G|36°C (D), CVS|N2cG|36°C (E), and CVS|A-N2cG|36°C (F). The schematic drawing of the advanced helper plasmid (*UGNT-A*) is shown in (F). Dashed yellow circles, starter PCs; dashed white lines, the cerebellar boundaries. L, lateral; R, rostral. Scale bars, 20 μ m.

(C and G) Summary of CI for the connection of traced neurons, glia or all cells to starter PCs at 2, 4, 6 and 10 dpi under the first two tracing conditions at 28°C (C) and the last three tracing conditions at 36°C (G). Significant intergroup differences exist in CI for traced neurons, glia or all cells (see the *P* values). There are no statistical differences between 6 and 10 dpi in any of the groups (n.s.). Two-way ANOVA and nonparametric two-tailed Mann-Whitney test for intergroup and intragroup statistics, respectively. Error bars denote SEM. Numbers in brackets indicate the number of larvae examined.

(H and I) Boxplots of CI for the connection of traced glia (H) or all cells (I) to starter PCs under six tracing conditions at 6 - 10 dpi. See also **Figures 2C–2E**. Center, median; bounds of box, first and third quartiles; whiskers, minimum and maximum values. n.s., not-significant; **P* < 0.05; ***P* < 0.01; ****P* < 0.001 (nonparametric two-tailed Mann-Whitney test).

(dpf). The labeling of input neurons usually emerged around 2 days post injection (dpi), and all CIs were calculated at 6 - 10 dpi, during which the viral transfer was relatively stable (**Figure 2—figure supplement 1C**).

After elevating the temperature from 28°C to 36°C for rearing virus-injected larvae, the neuron tracing efficiency of SADdG[EnvA] became obvious (**Figure 2C,D**, **Figure 2—figure supplement 1D**, and **Figure 1,2—table supplement 1**; CI_{neuron} : 4.96 ± 1.01 vs 0.06 ± 0.03 , $P < 0.001$); for CVSdG[EnvA] trans-complemented with N2cG, the efficiency further increased by one order of magnitude (**Figure 2C,D**, **Figure 2—figure supplement 1E** and **Figure 1,2—table supplement 1**; CI_{neuron} : 11.61 ± 2.43 vs 1.02 ± 0.18 , $P < 0.001$). In addition, the elevated temperature also significantly increased the proportion of initially infected larvae (**Figure 2F**) and the tendency of viral spread to neurons (**Figure 2E**). This efficient trans-synaptic viral transfer from PCs allowed clear labeling of GC parallel fibers and long-range contralateral inputs from IOCs (**Figure 2G-I**). We then generated a *Tetoff* element-based advanced helper plasmid (UGNT-A) to enhance the expression level of CVS N2cG (A-N2cG) (**Figure 2—figure supplement 1F**, see Methods), and this optimization resulted in a nearly two-fold increase in the neuron tracing efficiency (**Figure 2C,D**, **Figures 2—figure supplement 1F,G** and **Figure 1,2—table supplement 1**; CI_{neuron} : 20.35 ± 3.45 vs 11.61 ± 2.43 , $P < 0.05$). The improvement was also observed for tracing glia (**Figure 2C**, **Figure 2—figure supplement 1H,I** and **Figure 1,2—table supplement 1**).

To further functionally test the synaptic specificity of RV retrograde spread in the cerebellar circuit, we conducted *in vivo* Ca^{2+} imaging of starter PCs while simultaneously activating single traced GCs via electrical stimulation (**Figure 2J**, see Methods). To monitor the activity of starter PCs, the fluorescent protein in the helper plasmid was replaced with the Ca^{2+} indicator GCaMP6s (UG6sNT-A, Methods). GC stimulation-induced Ca^{2+} activities in starter PCs were abolished after two-photon laser-based ablation of the GC stimulated (**Figure 2K,L**). These results indicate that RV can retrogradely spread to GCs which are functionally connected with starter PCs, confirming the capability of RV to trace monosynaptic inputs in larval zebrafish.

Time window for normal neuronal health after RV infection

In sum, we have identified the optimal conditions for efficient retrograde monosynaptic tracing in zebrafish larvae: CVSdG[EnvA] trans-complemented with A-N2cG via the GAL4-UAS system at 36°C. The high efficiency of these optimal tracing conditions prompted concerns regarding the potential cytotoxicity of CVSdG and CVS N2cG. In line with previous reports in mice²⁰, we found lower toxicity of CVSdG compared with SADdG. At 36°C, the survival rate of larvae infected with CVSdG was significantly higher than those infected with SADdG (**Figure 3A**; $75\% \pm 3\%$ vs $52\% \pm 7\%$, $P < 0.05$). Furthermore, by using the time-lapse imaging as shown in Figure S2, we often observed that early emerged starter cells disappeared in larvae infected with SADdG (**Figure 3B**). As new starter cells continued to appear from 2 to 6 dpi (**Figure 3C**), we utilized starter cells emerged at 2 dpi to quantify their lifetime. Consistently, starter cells infected by CVSdG survived longer than those infected by SADdG (**Figure 3D**; 10.0 ± 0 days vs 7.5 ± 0.48 days, $P < 0.001$). Notably, the viability of starter cells decreased when CVSdG was trans-complemented with A-N2cG (**Figure 3D**; 10.0 ± 0 days vs 8.7 ± 0.48 days, $P < 0.001$).

To further test whether physiological functions of starter cells were impaired by CVSdG trans-complemented with A-N2cG, we conducted *in vivo* whole-cell recordings on starter PCs at around 10 dpi (**Figure 3E**, see Methods). Infected PCs exhibited normal spontaneous excitatory postsynaptic currents and potentials (sEPSCs and sEPSPs), and current injection-evoked firing patterns (**Figures 3F,G**)⁴⁰. Then we examined Ca^{2+} activities of infected (i.e., starter cells) and uninfected PCs (i.e., non-starter cells) in respond to complex visual stimulation⁴¹ (see Methods). No significant differences were detected between starter and non-starter PCs (**Figure 3H-L**).

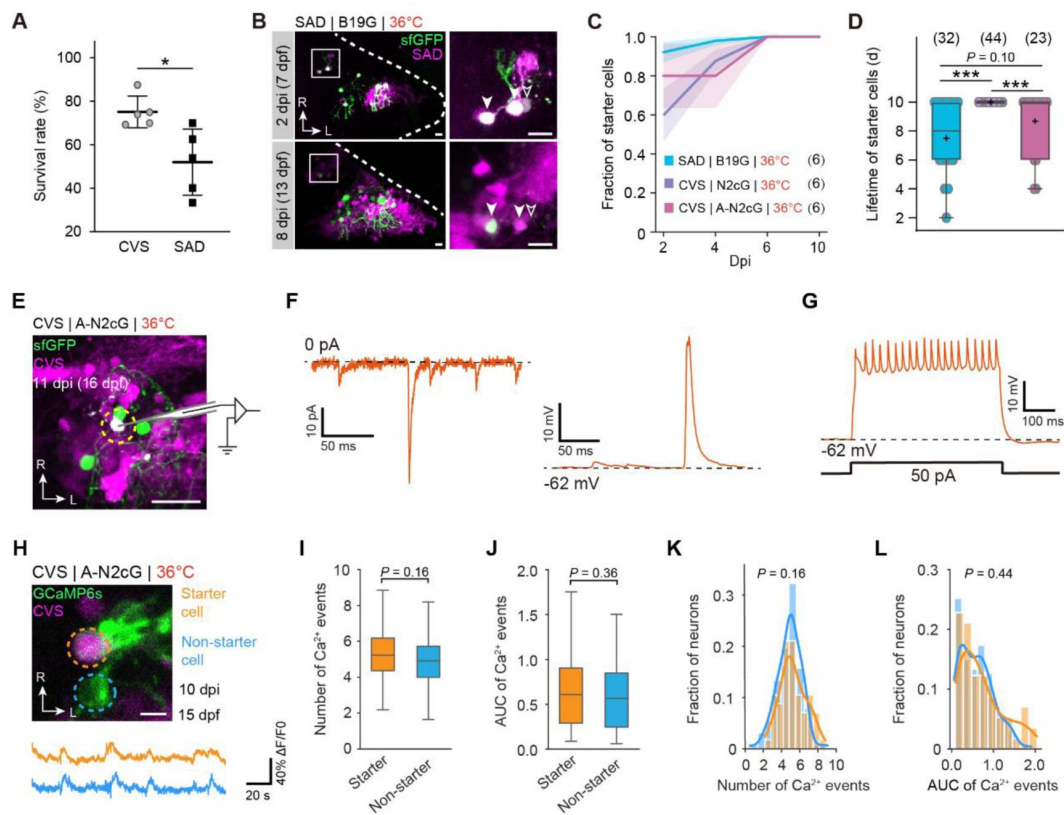


Figure 3.

Neuronal health and physiology appeared normal for at least 10 days after RV infection

(A) Survival rate of larvae injected with SADdG-mCherry[EnvA] or CVSdG-tdTomato[EnvA]. Five independent experiments were repeated.

(B) Time-lapse confocal images showing a starter cell (unfilled white arrowhead) present at 2 dpi disappears by 8 dpi, and two starter cells (white arrowheads) that remain present from 2 to 8 dpi. Boxed regions are enlarged on the right. Dashed lines indicate the cerebellar boundaries.

(C) Cumulative proportion of starter cells that have appeared at 2, 4, 6 and 10 dpi to the total number of observed starter cells under the three tracing conditions, showing the continuous appearing of starter cells at 2 - 6 dpi. Numbers in brackets indicate the number of larvae examined. The shadowed area represents SEM.

(D) Summary of the lifetime of starter cells appeared at 2 dpi under the three tracing conditions. The color scheme follows the same conventions shown in (C). Numbers in brackets indicate the number of starter cells examined.

(E) Schematic illustration of *in vivo* whole-cell recording on a starter PC at 11 dpi.

(F) Example traces of spontaneous EPSCs (left) and EPSPs (right) from starter PCs.

(G) Example trace of spikes in response to current-step injection into a starter PC.

(H) Confocal image (top) and Ca^{2+} responses (bottom) of a starter PC (dashed orange circle) and non-starter PC (dashed blue circle) to visual stimuli at 10 dpi.

(I-L) Boxplots (I and J) and distributions (K and L) of Ca^{2+} activity event number (I and K) and area under curve (AUC) of Ca^{2+} activity events (J and L) of starter PCs ($n = 88$) and non-starter PCs ($n = 127$) in larvae ($N = 8$) at 6 - 10 dpi.

Scale bars, 5 μm (H), 10 μm (B), 20 μm (E). SAD/ CVS | B19G/ N2cG/ A-N2cG | 28/36°C, the tracing conditions written as the G-deleted RV strain that was injected | the helper glycoprotein used for trans-complementation | fish rearing temperature after virus injection. L, lateral; R, rostral. Boxplots in (D, I and J): center, median; cross symbol, mean; bounds of box, first and third quartiles; whiskers, minimum and maximum values. n.s., not-significant; * $P < 0.05$; *** $P < 0.001$ (two-tailed unpaired Student's *t* test). Error bars denote SEM.

Taken together, these results indicate that under the optimal tracing conditions, infected neurons can remain relatively healthy up to 10 days after infection in larval zebrafish, allowing for functional probing of traced neural circuits.

Input cell-type-specific tracing and circuit reconstruction in the cerebellum

One important application of neural circuit tracing is to map the structural connectivity of specific neuron types to build the brain connectome atlas. As shown above, even under the optimal tracing conditions, glia still constituted around 47% on average of the traced cells (see **Figure 2C,E**). This will certainly affect discrimination of traced neurons and their neural fibers. To confine retrograde tracing to neurons instead of glia, we generated a Cre-Switch⁴² transgenic reporter fish *Tg(elavl3:Tetoff-DO_DIO-Hsa.H2B-mTagBFP2_tdTomato-CAAX)*. This reporter expresses pan-neuronally nuclear-localized blue fluorescent protein (mTagBFP2) by default, which serves as a reference for image registration and data integration. It also achieves Cre-dependent neuron-specific labeling with tdTomato-CAAX via the *elavl3* promoter (**Figure 4A**, see Methods). We employed this reporter together with Cre-expressing CVSdG (CVSdG-Cre[EnvA]) to map neuronal inputs to PCs in the cerebellum (**Figure 4B**). The transgenic labeling of traced neurons showed much brighter and sharper fluorescence compared with virus-assisted labeling (**Figure 4C**). Among all 13 fish examined, we detected tdTomato-CAAX signals exclusively in neurons without any labeling of glia (**Figure 4C**), facilitating the reconstruction of traced neural circuits (**Figure 4D**). We noticed a relatively lower tracing efficiency of this Cre-dependent tracing system (**Figures 4C, 2C**; $CI_{\text{neuron}}: 5.20 \pm 1.09$ vs 11.61 ± 2.43). Ongoing efforts are performed to increase the expression completeness of the transgenic reporter and improve the efficiency of Cre-mediated recombination.

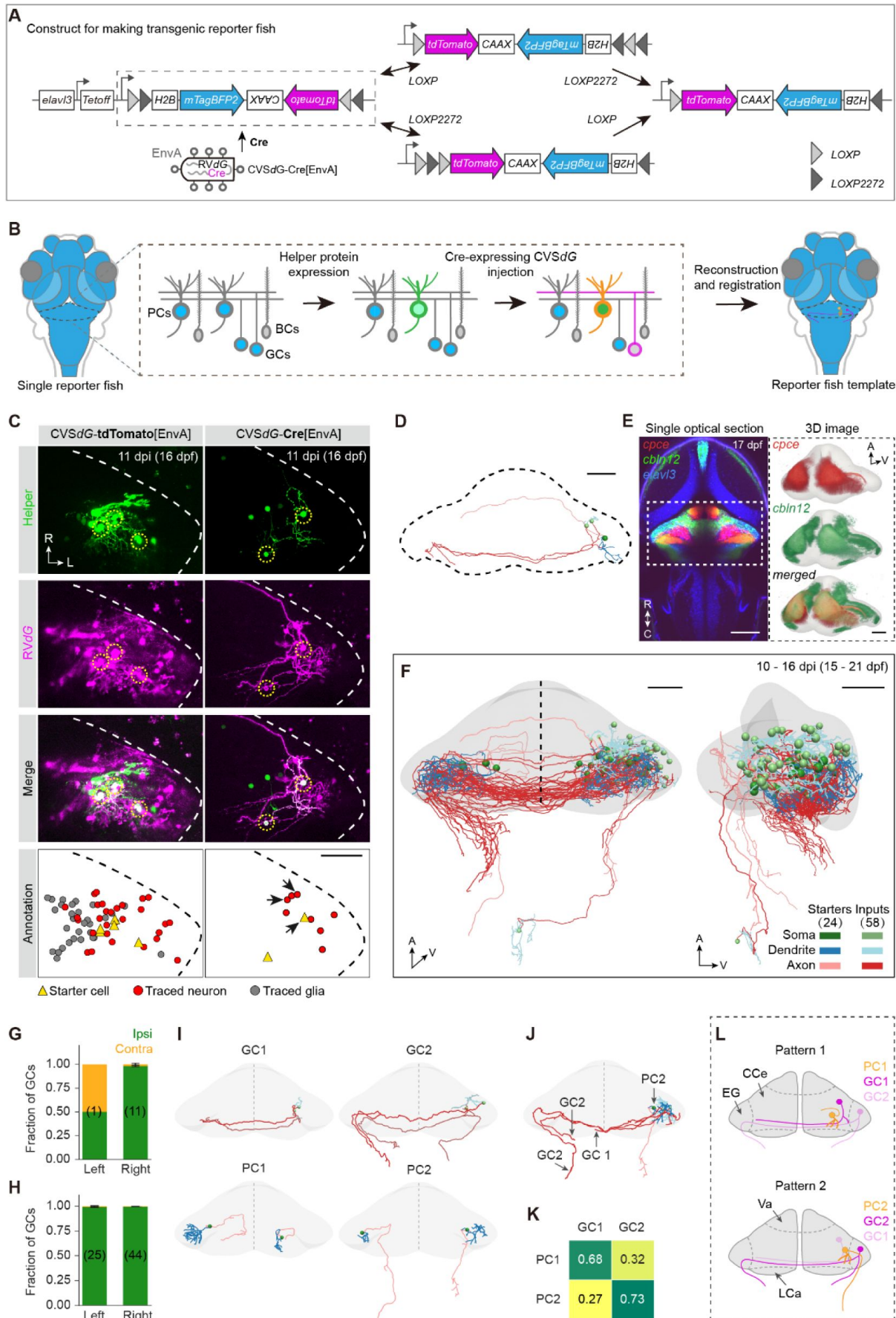


Figure 4.

Input cell-type-specific tracing and circuit reconstruction in the cerebellum

(A) A single Cre-Switch⁴² transgene (*elavl3:Tetoff-DQ_DIO-Hsa.H2B-mTagBFP2_tdTomato-CAAX*; abbr., *elavl3DoDioBR*) was generated to achieve a reference transgene (*elavl3:H2B-mTagBFP2*) for image registration and the Cre-dependent reporter expression (*tdTomato-CAAX*) specifically in neurons (*elavl3*⁺). Two ORFs flanked by two pairs of LOX sites oriented oppositely with the activation of forward-oriented *H2B-mTagBFP2* ORF by the *elavl3* promoter in global neurons. After Cre-expressing virus infection, Cre acts on either pair of LOX sites (LOXP (gray triangles) or LOXP2272 (black triangles)), resulting in stable inversion of the two ORFs and the activation of *tdTomato-CAAX* ORF by *elavl3* promoter in initially infected starter neurons and transneuronally infected input neurons.

(B) Schematic of building PC input atlas in larval zebrafish. The *elavl3DoDioBR* reporter fish (pan-neuronal H2B-mTagBFP2⁺) expressing helper proteins in PCs (sfGFP⁺/mTagBFP2⁺) were injected with Cre-expressing CVSdG[EnvA], which were trans-complemented in starter PCs and spread to direct inputs, resulting in Cre-dependent expression of the reporter protein (tdTomato-CAAX) in *elavl3*⁺ starter PCs (sfGFP⁺/tdTomato⁺/mTagBFP2⁺) and input GCs (sfGFP⁺/tdTomato⁺/mTagBFP2⁺). The pan-neuronally-expressing *H2B-mTagBFP2* in *elavl3DoDioBR* serves as the reference and bridge for the registration of reconstructed circuits. BCs, Bergmann glia; GCs, granule cells; PCs, Purkinje cells.

(C) Example of *in situ* complementation (11 dpi) in WT larvae (left) or *elavl3DoDioBR* reporter larvae (right) infected with CVSdG that was trans-complemented with CVS N2cG at 36°C, showing exclusive labeling of neurons in tdTomato⁺-only traced cells when using Cre-dependent *elavl3DoDioBR* reporter fish. Confocal images (top) and corresponding schematic illustrations (bottom) are shown. Dashed yellow circles, starter PCs; dashed lines, the cerebellar boundaries. Black arrows indicate the reconstructed neurons shown in (D).

(D) Dorsal 3D view of the reconstructed starter PC and two input GCs. Dashed line indicates the cerebellar boundary. The color scheme follows the same conventions as in (F).

(E) Left, single horizontal section view of *Tg(2×en.cpce-E1B:tdTomato-CAAX)* (red), *Tg(cbln12:GAL4FF);Tg(5×UAS:EGFP)* (green), and *elavl3DoDioBR* (blue) templates aligned in a common coordinate space. Right, volume rendering of the cerebellar area (dashed white rectangle on the left) of the templates labeling PCs (red) and GCs (green) in outline of cerebellum region (gray).

(F) Dorsal (left) and lateral (right) 3D view of reconstructed starter PCs (n = 24) and input GCs (n = 58) in the cerebellum (gray). Different subcellular parts of PCs and GCs are color-coded. Data were collected from 13 larvae at 10 - 16 dpi (15 - 21 dpf).

(G and H) Ipsilateral vs contralateral proportions of GC inputs to PCs in the left and right cerebellum. Analysis were performed using the reconstructed data in (E) and the viral tracing data shown in **Figures 2C** and **2D** for (G) and (H), respectively. Numbers in brackets indicate the number of larvae examined which only had starter PCs on one cerebellar hemisphere.

(I) Subtypes of reconstructed PCs and GCs. Two typical cells are shown for each subtype.

(J) Example showing three GCs traced from a single PC.

(K) Summed input proportion of two subtypes of GCs to two subtypes of PCs. Cases with only one subtype of PC and clearly identifiable GCs were included. The cell numbers are as follows: PC1, 12; PC2, 6; GC1, 19; GC2, 18.

(L) Schematic representation of subtype-specific connectivity patterns from GCs to PCs. High saturation of color indicates a higher input strength. Cc, corpus cerebelli; EG, eminentia granularis; LCa, lobus caudalis cerebelli; Va, valvula cerebelli.

Scale bars, 50 μ m. A, anterior; C, caudal; L, lateral; R, rostral; V, ventral. Dashed vertical lines indicate the midline of the cerebellum.

To demonstrate the feasibility of our tracing system in building and analyzing brain input maps, we then generated a three-dimensional (3D) reference template brain from the transgenic reporter fish at 17 dpf (see Methods). To incorporate cerebellar expression patterns, we generated the transgenic lines *Tg(2 $\times$ en.cpce:tdTomato-CAAX)* and *Tg(cbln12:GAL4FF)* to label PCs and GCs, respectively. By utilizing H2B-mTagBFP2 expression in the reporter fish as a bridge, we mapped the two cerebellar expression patterns and circuit tracing results of 13 fish mentioned above onto the common coordinate space of the reference template (**Figure 4B,E**, see Methods). We delineated the cerebellar brain area and reconstructed each starter PC ($n = 24$) and its input GCs ($n = 58$) with clearly visible morphology (**Figure 4E,F**), resulting in a digital cellular-resolution input atlas of cerebellar PCs (**Figure 4F**).

This input atlas clearly shows an ipsilateral preference of afferents from GCs to PCs (**Figure 4G**), which was in accordance with the summarized results of non-Cre-dependent viral tracing data (**Figures 4H** and **2C,D**). Interestingly, all reconstructed GCs also innervated contralaterally by crossing the midline. We found two morphological subtypes for both GCs and PCs in the reconstruction, and each included one local subtype with axonal projections within the cerebellar area and one long projection subtype with axons extending caudally to the dorsal hindbrain^{43,44} (**Figure 4I**). Interestingly, although a single PC can receive inputs from the two subtypes of GCs (**Figure 4J**), the two subtypes of PCs prefer inputs from different subtypes of GCs (**Figure 4K,L**). These results demonstrate how cellular-level connectivity can advance our understanding of circuit wiring patterns.

Together, the development of the Cre-dependent tracing tool empowers the RVdG-based circuit tracing method, enabling input-specific tracing, cellular-level circuit reconstruction, and integrated mapping and analysis of afferent connections in larval zebrafish.

Discussion

Understanding brain functions requires systematic dissection of the involved neural circuits. Here, we developed a highly feasible and efficient implementation of the viral tracing tool RVdG[EnvA] for circuit studies in larval zebrafish (**Figure 1–table supplement 1**). Through step-by-step improvement, we identified an optimal tracing condition for larval zebrafish, which includes the CVSdG strain, the native G protein of the CVS (i.e., N2cG), an enhanced expression system for the G protein, and elevated rearing temperature. Our method exhibited a 20-fold increase in tracing efficiency compared to previous studies in adult zebrafish^{26,27}, and demonstrated low cytotoxicity. Moreover, we established a custom-designed transgenic reporter framework, which enables genetic access to specific input cell types and allows for cellular-resolution reconstruction and alignment of traced circuits. We demonstrated the application of this framework to decipher cell-type specific connections from GCs to PCs in the cerebellum.

Helper protein expression design that guarantees high feasibility of the tracing method

Well-controlled spatial expression of helper proteins (i.e. TVA and G) in starter cells is critical for the application of RVdG[EnvA]-based circuit tracing. Previous studies on zebrafish expressed TVA and G through generating transgenic fish lines^{26,27} or injecting helper viral vectors^{27,29,30}. Both approaches allow for genetic access to a specific types or groups of

	This work	Dohaku <i>et al.</i> 2019 ²⁶	Satou <i>et al.</i> 2022 ²⁷		Ma <i>et al.</i> 2020 ²⁹	Kler <i>et al.</i> 2021 ³⁰
Age	3 - 21 dpf	Adult	Adult	7 - 14 dpf	2.5 - 9 dpf	3 - 9 dpf
Region being studied	Brain	Brain	Brain	Spinal cord	Brain	Brain
Virus strains	RV-SADd G[EnvA]; RV-CVSd G[EnvA]	RV-SADd G[EnvA]	RV-SADd G[EnvA]	RV-SADd G[EnvA]	rVSVd G[VSVG]	rVSV(M51R)d G[VSVG]
Glycoprotein	SAD-B19G; CVS-N2cG	SAD-B19G	SAD-B19G	SAD-B19G	VSVG	VSVG
Spread direction	Retrograde	Retrograde	Retrograde	Retrograde	Anterograde	Anterograde
Highest Tracing efficiency (CI)	~20	N.A.	~1	N.A.	N.A.	N.A.
Genetic targeting of starter cells?	Y (EnvA-TVA)	Y (EnvA-TVA)	Y (EnvA-TVA)	Y (EnvA-TVA)	N	N
Methods for helper proteins expression	Transient expression using single plasmid	Transgenesis	HSV1	HSV1	Lentivirus	Lentivirus
Co-expression of TVA & G?	Y	N	N	N	N	N
Synaptic specificity verification?	Y	N	N	N	N	N
Input or output selectivity?	Y	N	N	N	N	Y
Neuron viability test?	Y	N	Y	N	N	N
Life span of infected cells (d)	10 - 14 (starter cells)	N.A.	~6 (starter cells)	N.A.	<6 (traced cells)	<6 (traced cells)
Rearing temperature for virus-injected fish (°C)	28 & 36	~35	36	36	28	28
Experimental procedure feasibility ■ 1.Helper protein expression ■ Plasmid microinjection ■ Transgenic fish ■ Helper virus ■ 2.Virus injection ■ 3.Fluorescence signal amplification ■ Brain slicing ■ Immunohistochemistry ■ 4.Imaging	■ ■ ■	■ ■ ■ ■ ■	■ ■ ■ ■ ■	■ ■ ■ ■ ■	■ ■ ■	■ ■ ■

Y, yes; N, no; N.A., not available

Figure 1-4—table supplement 1.

Properties of currently available viral circuit tracing systems for zebrafish.

neurons through the GAL4-UAS system. The transgene typically results in expression in the majority of neurons of the targeted type that usually distribute in many brain areas. Virus-assisted expression depends on the distribution territory of injected virus. Restricting to specific sub-regions of the brain is quite challenging due to the small compact brain of zebrafish larvae. Our method employs the transient transgene technique in zebrafish to express helper proteins. This involves temporarily introducing GAL4-UAS binary DNA plasmids into the fish's genome through microinjection into fertilized eggs (see **Figures 1A** and **2A**). It can achieve sparse expression of TVA and G in genetically defined cell types, even in a single neuron. Tracing from a single neuron can help cell-subtype-specific circuit dissection. Furthermore, we linked the two helper protein genes with the 2A element to co-express TVA and G in the same neuron (see **Figures 1A** and **2A**). Co-expression guarantees that the tracing is monosynaptically restricted. This is because if G-expressing-only neurons exist, they may act as input cells for initially infected starter cells, resulting in further trans-complementation and spread of RVdG across the second and even more synaptic steps³². Previous studies on zebrafish expressed TVA and G separately using either two transgenic lines or two viral vectors^{26,27}, potentially resulting in separate expression of them in different neurons.

Our study demonstrated that the transient expression of helper proteins effectively facilitates subsequent viral infection and trans-complementation, offering a time-efficient and cost-effective alternative to generating stable transgenic lines. It also addresses the potential cytotoxicity effects associated with the use of helper viral vectors.

With commercially available recombinant RV tools, our method can be easily implemented by zebrafish research laboratories using three commonly used techniques: plasmid construction, microinjection (DNA constructs and recombinant RV with a 3 - 6 day interval) and live optical imaging. Additional enhancements for the helper plasmid could include the implementation of light-responsive inducible systems⁴⁵ to achieve both temporal and spatial control of helper protein expression.

Key factors that influence tracing efficiency

To enhance RVdG tracing efficiency, we introduced the CVS-N2c rabies strain in zebrafish for the first time. It has been reported that the CVS-N2c strain exhibits significantly higher tracing efficiency in mice compared to the SAD-B19 strain, with an improvement of almost tenfold²⁰. In zebrafish larvae, CVSdG, when trans-complemented with its native N2cG, also showed an average increase close to tenfold compared to SADdG trans-complemented with its native B19G (see **Figure 2D** and **Figure 1,2—table supplement 1**, CI_{neuron}). This improvement may be attributed to the reduced replication and protein expression of the CVS-N2c²⁰. This could result in decreased toxicity to host cells, longer reproductive times, and ultimately higher spread efficiency. Consistently, in zebrafish larvae, we found that neurons infected with CVS-N2c displayed lower fluorescence intensity and longer survival times compared to those infected with SAD-B19. Notably, N2cG exhibited higher tracing efficiency than B19G when they were used to trans-complement the same CVSdG (see **Figure 2D** and **Figure 1,2—table supplement 1**). This could be attributed to the greater neurotropism of N2cG compared to B19G, or potential mismatches between the cytoplasmic domain of B19G and the CVS-N2c capsid. Further investigation into the impact of different G proteins on RVdG tracing efficiency in zebrafish would benefit from the construction of chimeric G proteins for comparison⁴⁶.

Besides the virus strain, temperature also plays a pivotal role. Under the optimal temperature for virus replication at 36°C, the tracing efficiency of CVS-N2c increased more than tenfold in comparison with that under the standard rearing temperature for zebrafish at 28°C. This effect was more pronounced for SAD-B19, showing an increase of over 80 times (see **Figure 2D** and **Figure 1,2—table supplement 1**). Importantly, in line with previous reports²⁷, we did not observe any noteworthy increase in mortality or behavioral abnormalities among zebrafish larvae exposed to elevated temperatures.

The expression level of G proteins is another determinant for efficient rabies tracing⁴⁶. Building upon the codon optimization of the G protein, we further increased the expression of N2cG by inserting the Tetoff element upstream of the N2cG gene (A-N2cG) in the UAS helper plasmid (see **Figure 2-figure supplement 1F**, **Figure 2D** and **Figure 1,2—table supplement 1**). This integration enables dual amplification of N2cG and TVA expression by both the GAL4-UAS and tTA-TRE binary systems. Combining the three key optimal conditions: CVS-N2c strain, 36°C temperature, and A-N2cG, we have achieved the highest tracing efficiency in zebrafish to date, with a 20-fold increase compared to previous reports²⁷.

Monosynaptic specificity of the tracing

The RVdG-based transneuronal labeling of radial glial cells was commonly observed in larval zebrafish^{29,30} (see **Figure 1D,G** and **Figure 2—figure supplement 1H**). The viral spread to glial cells may cause concerns about the synaptic specificity of RVdG spread between neurons. By employing the cerebellar circuit, we examined the complete pattern of traced neurons from PCs in larval zebrafish, and found that CVSdG was able to trace well-known presynaptic neurons of PCs, including both intra-cerebellar GCs and extra-cerebellar IOCs (see **Figure 2G-I**). Importantly, IOCs were the only labeled cells outside of the cerebellum, demonstrating the retrograde and monosynaptic specificity of this viral tracer in the zebrafish brain. We further provided functional evidence that CVSdG, even at its highest efficiency with A-N2cG, spreads retrogradely between neurons through synaptic connections (see **Figure 2H-L**). In all experiments involving the successful activation of PCs in response to electrical stimulation of single GCs, we consistently did not observe PC activation after ablating the GCs.

Low cytotoxicity that enables functional experiments in a wide time window

We found that the CVS-N2c strain did not significantly affect the neuronal health and physiology of starter cells for at least 10 days after infection (see **Figure 3D-L**). Meanwhile, consistent with previous reports in mice²⁰, the SAD-B19 strain exhibited higher cytotoxicity than CVS-N2c, resulting in more cell death of starter cells (see **Figure 3D**). It indicates that CVS-N2c is more suitable for application in zebrafish. We could observe the labeling of starter cells even one month later after CVS-N2c infection, and the infected zebrafish could be raised to adulthood. It is worth noting that enhanced G protein expression achieved through the introduction of the Tetoff element (i.e. A-N2cG) in the helper plasmid improved tracing efficiency, but also resulted in elevated levels of cytotoxicity, though lower than that induced by SAD-B19 (see **Figure 3D**). This indicates an association between the expression levels of G proteins and cytotoxicity^{47,48}. Given the transferred RV is deficient in G, it is expected that retrogradely infected input cells will have lower cytotoxicity levels than starter cells. Therefore, input cells will survive for longer periods of time. Based on our experience, the CVS-N2c can be microinjected into zebrafish larvae between 3 and 6 dpf, and stable and viable labeling of traced circuits can be observed within 6 to 16 dpi. Therefore, the observation time window ranges from 9 to 22 dpf, spanning the larval ages commonly used for whole-brain functional imaging studies^{14,49}.

Functions and expandability of the transgenic framework

In the transgenic reporter framework (i.e. *elavl3DoDioBR*) (see **Figure 4A**), we designed three expansion motifs: 1) a promoter (*elavl3*) motif that allows genetic access to the expression of reporter cassettes in specific cell types, e.g. the *vglut2a* and *gad1b* gene promoters to restrict excitatory and inhibitory targeting, respectively; 2) a Cre-dependent “Switch-On” reporter (tdTomato-CAAX) motif that allows the expression of various structural and functional tools in traced neurons after viral spread, such as photoconvertible fluorescent proteins and optogenetic tools; 3) a Cre-dependent “Switch-Off” reporter (H2B-mTagBFP2) motif that allows a default

reporter to be turned off specifically in infected neurons. In addition, the helper plasmid can serve as additional expansion motif that works together with the Cre-dependent transgenic reporter, allowing for full genetic visualization and dissection of neural circuits¹⁵.

In this study, we used the *elavl3* promoter to restrict the fluorescence reporter to neurons and the “Switch-On” reporter tdTomato-CAAX to label the membrane of traced neurons. This enables clear neuron morphology labeling (see **Figure 4C**), facilitating the reconstruction of traced neural circuits based on single-neuron morphology. We used H2B-mTagBFP2 as the default “Switch-Off” reporter to provide a reference expression pattern for registering and integrating traced circuit images from different individual fish and experiments. As an application demonstration, we applied the transgenic framework to map the monosynaptic inputs to PCs from GCs in the cerebellum of larval zebrafish at ~ 2.5 weeks old (see **Figure 4**). Interestingly, we uncovered two properties of the cerebellar circuit: 1) the ipsilateral preference of GC-to-PC connections; 2) the subtype specificity of GC-to-PC connections. Further investigation is required to elucidate the functional implications of subtype-specific GC-to-PC connections and the apparent contradiction between the bilateral projection of GC parallel fibers and the GCs’ preference for ipsilateral connections to PCs. The reference expression pattern can serve as a reference atlas to integrate other modal data from brain-wide images with information on gene expression, neurochemistry and neuronal activity. This will largely enhance our understanding of neural wiring patterns in relation to other neural characteristics⁵⁰.

Conclusion

Our study established an applicable and effective viral circuit tracing system for larval zebrafish. We believe that this tracing system will offer an avenue for the integrated structural and functional dissection of neural circuits underlying a wide range of brain functions in larval zebrafish.

Methods

Animals

Adult zebrafish (*Danio rerio*) were reared under standard laboratory conditions, with temperatures set at 28°C, and a light/dark cycle of 14 hr/10 hr. The larval zebrafish used were in *nacre* background and at 3 - 21 dpf. The sex of the larvae at this stage is not determined. All experimental procedures were approved by the Animal Care and Use Committee of the Center for Excellence in Brain Science and Intelligence Technology (CEBSIT), Chinese Academy of Sciences (NA-046-2023).

Generation of DNA constructs and transgenic lines

The GAL4-UAS binary plasmids were applied to express fluorescent protein, TVA protein, and viral G in specific neuronal types. To randomly target neurons, we used *elavl3:GAL4-VP16*⁵¹. To target Purkinje cells (PCs), we generated a construct to drive the *GAL4FF* under the control of *cpce* (*ca8* promoter derived PC-specific enhancer element) fused to *E1B*³⁹. The *cpce* sequence was PCR-amplified from the zebrafish genome DNA, and the *cpce-E1B:GAL4FF* was flanked by a minimal *ToI2* transposase recognition sites⁵². We generated seven helper plasmids (abbreviated names are in parentheses): *14×UAS-E1B:EGFP-P2A-TVA* (*UGT*), *14×UAS-E1B:EGFP-P2A-TVA-T2A-B19G* (*UGTB*), *14×UAS-E1B/5×UAS-hsp:EGFP-P2A-B19G-T2A-TVA* (*UGBT*, the *14×* and *5×UGBT* were used with *elavl3:GAL4-VP16* and *cpce-E1B:GAL4FF*, respectively), *5×UAS-hsp:sfGFP-P2A-N2cG-P2A-TVA* (*UGNT*), *5×UAS-hsp:sfGFP-P2A-Tetoff-N2cG-P2A-TVA* (*UGNT-A*), and *5×UAS-hsp:GCaMP6s-P2A-Tetoff-N2cG-P2A-TVA* (*UG6sNT-A*). The *sfGFP*, *N2cG*, *B19G*, *TVA*, and *Tetoff* were codon optimized

for zebrafish and *de novo* synthesized; the *GCaMP6s* was PCR-amplified from *elavl3:GCaMP6s* plasmid (gift from M. B. Ahrens, Janelia Research Campus, USA). All these elements were sequentially assembled with 2A sequences and the linearized *miniTol2-UAS* backbone to generate polycistronic vectors through In-Fusion cloning. Sparse expression of helper proteins was achieved by the microinjection of a mixture (1 nl) of *elavl3* or *cpce*-driven GAL4 plasmid, UAS-driven helper plasmid, and *Tol2* transposase mRNA into one-cell stage zebrafish embryos at a concentration of 30 (10+10+10) ng/μl using an air-puffed pressure injector (MPPI-3, ASI).

To generate *Tg(2×en.cpce-E1B:tdTomato-CAAX)*, we created a construct to drive *tdTomato* with a 3' membrane tag encoding the CAAX box of human Harvey Ras (CTGAACCTCCTGATGAGAGTGGCCCCGGCTGCATGAGCTGCAAGTGTGTGCTCTCC) under the control of two tandem *cpce* fused to E1B. To generate *Tg(cbln12:GAL4FF)* for labeling granule cells (GCs) in the cerebellum, we amplified a ~2-kbp regulatory element from zebrafish genome DNA by PCR for the *cbln12* (*cerebellin12*) gene and subcloned it into the linearized *miniTol2-GAL4FF* backbone. To generate *elavl3DoDioBR*, we first *de novo* synthesized a zebrafish codon-optimized *mTagBFP2* and constructed an intermediate plasmid containing *Tetoff-DO_DIO-Hsa.H2B-mTagBFP2_tdTomato-CAAX* using In-Fusion cloning. Subsequently, we subcloned this target fragment into the linearized *miniTol2-elavl3* backbone to generate the final vector. Transgenic zebrafish were prepared using the *Tol2* transposase-based approach. Purified target plasmid DNA was microinjected with the *Tol2* transposase mRNA into one-cell stage zebrafish embryos at a concentration of 60 (30+30) ng/μl. Injected embryos were raised to sexual maturity to identify founder fish. The *Tg(cbln12:GAL4FF)* was crossed with *Tg(5×UAS:EGFP)* to visualize GCs.

Production of EnvA-pseudotyped rabies virus

The EnvA-pseudotyped rabies viruses (RV), including SADdG-mCherry[EnvA], CVSdG-tdTomato[EnvA], and CVSdG-mCherry-2A-Cre[EnvA], were rescued and prepared following the previously reported method^{53,54}. Rabies vectors were stored at -80°C, and the titers were diluted to 2×10^8 infectious units per ml (IU/ml) with phosphate-buffered saline (PBS) before use. We found that mCherry fluorescence after CVSdG-Cre[EnvA] infection *in vivo* was much weaker than that of transgenetically expressed tdTomato-CAAX, and thus did not affect the observation and reconstruction of neuronal morphology.

Virus application

At 3 - 6 dpf, larvae expressing helper proteins in specific neurons were anesthetized using 0.03% tricaine methanesulfonate (MS-222) and mounted laterally in 1.0% low-melting agarose (Sigma) on a glass-bottom dish filled with the extracellular solution consisted of (in mM): 134 NaCl, 2.9 KCl, 2.1 CaCl₂•2H₂O, 1.2 MgCl₂•6H₂O, 10 HEPES, and 10 glucose (pH = 7.8, osmolality = 290 mOsm). Under an inverted fluorescence microscope, 2 - 5 nl of virus suspension was injected near the target cells (revealed by sfGFP or GCaMP6s fluorescence) in one cerebellar hemisphere using a microinjector (Nanoliter2010, WPI) moved with a motorized controller (MC1000e, Siskiyou). After injection, the fish were allowed to rest for one minute before withdrawing the pipette. Then, they were placed in standard tanks in an incubator set at temperatures of 28°C or 36°C.

Confocal and two-photon imaging

Larval fish were mounted dorsal-side up in 1.5% low-melting agarose on a custom-made chamber. Structural live imaging of virus-injected fish and transgenic lines was performed using an upright confocal laser scanning microscope (FV3000, Olympus) equipped with a 20× water-immersion objective (N.A., 0.9) at a 1-μm step size. For virus-injected fish, the imaging dimension was based on the specific region of viral infection and spread. For transgenic lines, the entire brain was imaged by multiple volumes, which were then stitched together using a custom ImageJ (<http://fiji.sc>) macro based on the Grid/Collection stitching plugin⁵⁵. Functional Ca²⁺ imaging of PCs

expressing GCaMP6s was performed using an upright two-photon microscope (FVMPE-RS, Olympus) equipped with a 25× water-immersion objective (N.A., 1.05) at a wavelength of 920 nm. The calcium activity was recorded at ~5 Hz on single optical plane.

Electrical stimulation of single granule cell

Within 6 to 10 days after virus injection, larvae were paralyzed with pancuronium dibromide (PCD, 1mM, Tocris 0693) for 1 min and then mounted in 1.5% low-melting agarose on a chamber filled with the extracellular solution. The agarose and skin over the caudal side of the cerebellum were removed for the access of stimulation electrodes. Stimulation microelectrodes (1 - 2 μ m tip opening) were created using borosilicate theta glass capillaries (BT-150-10, Sutter Instrument) through a micropipette puller (P-97, Sutter Instrument). Then a silver wire was inserted into the electrode filled with the extracellular solution. The electrode was positioned at and attached to the cell body of the target GC, which was identified by the fluorescence and morphology using confocal imaging. The stimulation was delivered by a flexible stimulus isolator (ISO-Flex, AMPI), consisting of 5 pulse trains (6 - 8 V, 10 ms duration for one pulse, 10 pulses, 50 Hz) with inter-trial intervals of 40 s. The electrical stimulation of single GC was synchronized with the Ca^{2+} imaging of PCs by a stimulator (Master-8, AMPI).

Two-photon laser-based ablation of granule cells

A two-photon laser (795 nm) was used to ablate the GC after electrical stimulation. Confirmation of a successful ablation involved the identification of a bulb-like structure in the bright-field image and the absence of the tdTomato fluorescence signal from the targeted GC body. The efficiency of the ablation was further validated one hour later, before performing post hoc electrical stimulation at the same position using the bulb-like structure and adjacent fluorescent cells as references.

In vivo whole-cell recording of Purkinje cells

Within 6 to 15 days after virus injection, larvae were paralyzed with α -bungarotoxin (1mM) for 1 min, then mounted in 1.5% low-melting agarose on a chamber filled with the extracellular solution. The agarose and skin over the caudal side of the cerebellum were removed for the access of recording pipettes. A recording micropipette (15 - 20 M Ω , 1 - 1.5 μ m tip opening) filled with internal solution was approached to the target PC by a motorized micromanipulator (MP-225, Sutter Instruments). The internal solution consisted of (in mM): 100 K-gluconate, 10 KCl, 2 $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, 2 $\text{Mg}_2 \cdot \text{ATP}$, 0.3 $\text{GTP} \cdot \text{Na}_4$, 2 phosphocreatine, 10 EGTA, and 10 HEPES (pH = 7.4, osmolality = 270 mOsm). The target PC was identified by double fluorescence (sfGFP⁺ and tdTomato⁺) using confocal imaging and was recorded at whole-cell mode with similar procedures as our previous work⁶. The EPSP recording and current-step delivery were conducted in current-clamp mode, and the EPSC was assessed in voltage-clamp mode. Data were acquired using a patch-clamp amplifier (MultiClamp 700B, Axon Instruments) and a digitizer (Digidata 1440A, Axon Instruments), and signals were sampled at 10 kHz with Clampex 10.6 software (Molecular Devices).

Visual stimulation

Larvae examined were not anesthetized and were fully embedded in low-melting agarose. Visual stimuli were generated by a custom program written in Python and presented using an LCD screen covered with red filter paper to avoid spectrum interference. The visual stimuli basically followed a prior study⁴¹. Briefly, three trials were presented and each trial consisted of three types of stimuli in the following order: (1) four-directional whole-field gratings in black and white (including three forward, one backward, one leftward, and one rightward directions); (2) windmill patterns in black and white rotating at 0.2 Hz and varying in velocity following a sine function (including six whole, two right-half, and two left-half filed stimuli presented in alternating

clockwise and counterclockwise directions); (3) whole-field flashes of 1-second duration (alternating between white and black, repeated 4 times). Except for the flash, each stimulus lasted 5 seconds, with a 5-second inter-stimulus interval.

Quantification of RVdG[EnvA]-based circuit tracing

Cells were counted by manually annotating the center of their cell bodies in image stacks using the Cell Counter plugin in ImageJ. This was done for cells showing green (sfGFP/GCaMP6s; i.e., infected and uninfected TVA-expressing cells), red (tdTomato/mCherry; i.e., traced input cells and starter cells), or yellow (both green and red; i.e., initially infected starter cells) fluorescence separately. To count the traced neurons and glia separately, we also manually identified and marked the neurons based on their significant morphological distinctions with glia in larval zebrafish. The number of traced glia was then calculated by subtracting the number of starter cells and traced neurons from the total number of cells with red fluorescence. To determine virus tracing efficiency, a convergence index (CI) was calculated as the number of traced inputs (neurons, glia, or all) divided by the number of starter cells.

Analysis of calcium imaging data

The acquired time-lapse images were aligned first, and individual regions of interest (ROIs) were manually circled on the averaged image. Each ROI corresponds to either a starter cell or a non-starter cell, and the mean grayscale value within each ROI over time was represented as F . In the electrical stimulation experiment, the baseline value F_0 was calculated as the average value of F from a 10-second interval before each stimulus. In the visual stimulation experiment, F_0 was calculated as the average value of F over a 30-second period preceding the stimuli. The relative change in the intensity of each ROI was calculated using the formula $(F - F_0)/F_0$. The area under the curve (AUC) and the number of Ca^{2+} activity events were determined by applying a threshold of $2 \times$ standard deviation (SD) of a 60-second baseline activity. All of the mentioned calculations were performed using custom Python scripts.

Template generation and registration

We first generated two two-channel templates (17 dpf), *Tg(2×en.cpce:tdTomato-CAAX);elavl3DoDioBR* and *Tg(cbln12:GAL4FF);Tg(5×UAS:EGFP);elavl3DoDioBR*, by running a shell script `antsMultivariateTemplateConstruction2.sh` (`-d 3 -b 1 -g 0.1 -i 10 -c 2 -j 6 -k 2 -w 1×1 -f 12×8×4×2 -s 4×3×2×1vox -q 400×200×200×20 -r 1 -n 0 -m CC -t SyN ${image-inputs}`) on the CEBSIT's Linux computing cluster using the open-source Advanced Normalization Tools (ANTs) software⁵⁶. Confocal images were acquired at $1 \times 1 \times 1 \mu\text{m}$ resolution. Both templates were shape-based average representations from 6 transgenic larvae after 7 iterations. Using normalized cross correlation (NCC) as a metric to evaluate the registration accuracy between *elavl3DoDioBR* templates and individual *elavl3DoDioBR* scans of larvae used for circuit tracing, we determined the *elavl3DoDioBR* in *Tg(2×en.cpce:tdTomato-CAAX);elavl3DoDioBR* template space as the common coordinate reference for registration. The expression pattern of *Tg(cbln12:GAL4FF);Tg(5×UAS:EGFP)* and circuit tracing results were transformed into the common coordinate space by using *elavl3DoDioBR* as a bridge. For image registration, we used `antsRegistration` and `antsApplyTransforms` functions in ANTs, following the registration parameters for live fish scans reported in previous work⁵⁷. Neuron reconstructions (SWC files⁵⁸) were aligned using `antsApplyTransformsToPoints` function in ANTs.

Cerebellum delineation and neuron reconstruction

The 3D cerebellum region was delineated manually using 3D Slicer software (<https://www.slicer.org/>) based on the expression patterns of *elavl3DoDioBR*, *Tg(2×en.cpce:tdTomato-CAAX)*, and *Tg(cbln12:GAL4FF);Tg(5×UAS:EGFP)* in the common coordinate space. The boundary delineation was performed on *elavl3DoDioBR*, assisted by overlaying the other two patterns.

To reconstruct the morphology of neurons in virus-traced circuits, neurons expressing membrane-bound tdTomato were semi-automatically traced using the simple neurite tracer (SNT)⁵⁹ plugin in ImageJ. The classification of dendritic and axonal compartments of neurites was based on structural knowledge. Neurons with low fluorescence intensity and ambiguous morphology were excluded during reconstruction. The resulting individual neuron skeletons were saved in SWC format. Tracings containing undistinguished several neuron skeletons were also saved as individual SWC files.

The co-volume rendering of template cerebellum and cerebellum region (see **Figure 4E**) was performed using ParaView⁶⁰ (<https://www.paraview.org>). The co-visualization of the cerebellar region and registered neuron reconstructions (see **Figure 4F,I,J**) was performed using custom python scripts based on NAVis package⁶¹ (<https://github.com/navis-org/navis>).

Statistics

All statistical tests and graphs were performed with GraphPad Prism or Python software. Data normality was first examined with Shapiro-Wilk test. Comparisons between two groups with normal and non-normal distributions were made using two-tailed unpaired Student's *t* test and nonparametric two-tailed Mann-Whitney test, respectively. Differences between groups involving multiple factors were analyzed using two-way ANOVA. *P* value < 0.05 achieved statistical significance. Data are represented as mean ± SEM. The sample size and/or the number of replicates for each experiment are reported in the text, figures or figure legends.

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Author contributions

X.F.D. and J.L.D. conceptualized and supervised the project. T.L.C. and Q.S.D. developed the methodology and designed the experiments with input from X.F.D. and J.L.D.. T.L.C. and Q.S.D. performed the experiments with input from X.W. and X.D.Z.. X.D.Z. created the cerebellum-related transgenic fish. K.Z.L. produced the rabies viruses under the supervision of F.Q.X.. T.L.C. analyzed data and made the illustrations with input from Q.S.D., Y.W.Z., and X.Y.N.. Y.L. helped to establish the virus injection in zebrafish. T.L.C., Q.S.D., X.F.D., and J.L.D. wrote the manuscript.

Data availability

Plasmids and transgenic zebrafish lines generated in this study, along with the microscopy and neuron reconstruction data reported in this paper, will be made available upon request. All original code has been deposited at <https://github.com/soaringdu/Proj-RVCT> and is publicly available as of the date of publication. Any additional information required to reanalyze the data reported in this paper is available upon request.

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Reviewer #1 (Public Review):

EnvA-pseudotyped glycoprotein-deleted rabies virus has emerged as an essential tool for tracing monosynaptic inputs to genetically defined neuron populations in the mammalian brain. Recently, in addition to the SAD B19 rabies virus strain first described by Callaway and colleagues in 2007, the CVS N2c rabies virus strain has become popular due to its low toxicity and high trans-synaptic transfer efficiency. However, despite its widespread use in the mammalian brain, particularly in mice, the application of this cell-type-specific monosynaptic rabies tracing system in zebrafish has been limited by low labeling efficiency and high toxicity. In this manuscript, the authors aimed to develop an efficient retrograde monosynaptic rabies-mediated circuit mapping tool for larval zebrafish. Given the translucent nature of larval zebrafish, whole-brain neuronal activities can be monitored, perturbed, and recorded over time. Introducing a robust circuit mapping tool for larval zebrafish would enable researchers to simultaneously investigate the structure and function of neural circuits, which would be of significant interest to the neural circuit research community. Furthermore, the ability to track rabies-labeled cells over time in the transparent brain could enhance our understanding of the trans-synaptic retrograde tracing mechanism of the rabies virus.

To establish an efficient rabies virus tracing system in the larval zebrafish brain, the authors conducted meticulous side-by-side experiments to determine the optimal combination of trans-expressed rabies G proteins, TVA receptors, and recombinant rabies virus strains. Consistent with observations in the mouse brain, the CVS N2c strain trans-complemented with N2cG was found to be superior to the SAD B19 combination, offering lower toxicity and higher efficiency in labeling presynaptic neurons. Additionally, the authors tested various temperatures for the larvae post-virus injection and identified 36°C as the optimal temperature for improved virus labeling. They then validated the system in the cerebellar circuits, noting evolutionary conservation in the cerebellar structure between zebrafish and mammals. The monosynaptic inputs to Purkinje cells from granule cells were neatly confirmed through ablation experiments.

However, there are a couple of issues that this study should address. Additionally, conducting some extra experiments could provide valuable information to the broader research field utilizing recombinant rabies viruses as retrograde tracers.

(1) It was observed that many radial glia were labeled, which casts doubt on the specificity of trans-synaptic spread between neurons. The issues of transneuronal labeling of glial cells should be addressed and discussed in more detail. In this manuscript, the authors used a transgenic zebrafish line carrying a neuron-specific Cre-dependent reporter and EnvA-CVS N2c(dG)-Cre virus to avoid the visualization of virally infected glial cells. However, this does

not solve the real issue of glial cell labeling and the possibility of a non-synaptic spread mechanism.

In addition, wrong citations in Line 307 were made when referring to previous studies discovering the same issue of RVdG-based transneuronal labeling radial glial cells.

"The RVdG-based transneuronal labeling of radial glial cells was commonly observed in larval zebrafish^{29,30}".

The cited work was conducted using vesicular stomatitis virus (VSV). A more thorough analysis and/or discussion on this topic should be included. Several key questions should be addressed:

- Does the number of labeled glial cells increase over time?
- Do they increase at the same rate over time as labeled neurons?
- Are the labeled glial cells only present around the injection site?
- Can the phenomenon of transneuronal labeling of radial glial cells be mitigated if the tracing is done in slightly older larvae?
- What is the survival rate of the infected glial cells over time?
- If an infected glial cell dies due to infection or gets ablated, does the rabies virus spread from the dead glial cells?
- If TVA and rabies G are delivered to glial cells, followed by rabies virus injection, will it lead to the infection of other glial cells or neurons?

Answers to any of these questions could greatly benefit the broader research community.

(2) The optimal virus tracing effect has to be achieved by raising the injected larvae at 36C. Since the routine temperature of zebrafish culture is around 28C, a more thorough characterization of the effect on the health of zebrafish should be conducted.

(3) Given the ability of time-lapse imaging of the infected larval zebrafish brain, the system can be taken advantage of to tackle important issues of rabies virus tracing tools.

a) Toxicity.

The toxicity of rabies viruses is an important issue that limits their application and affects the interpretation of traced circuits. For example, if a significant proportion of starter cells die before analysis, the traced presynaptic networks cannot be reliably assigned to a "defined" population of starter cells. In this manuscript, the authors did an excellent job of characterizing the effects of different rabies strains, G proteins derived from various strains, and levels of G protein expression on starter cell survival. However, an additional parameter that should be tested is the dose of rabies virus injection. The current method section states that all rabies virus preparations were diluted to 2×10^8 infection units per ml, and 2-5 nl of virus suspension was injected near the target cells. It would be interesting to know the impact of the dose/volume of virus injection on retrograde tracing efficiency and toxicity. Would higher titers of the virus lead to more efficient labeling but stronger toxicities? What would be the optimal dose/volume to balance efficiency and toxicity? Addressing these questions would provide valuable insights and help optimize the use of rabies viruses for circuit tracing.

b) Primary starters and secondary starters:

Given that the trans-expression of TVA and G is widespread, there is the possibility of coexistence of starter cells from the initial infection (primary starters) and starter cells generated by rabies virus spreading from the primary starters to presynaptic neurons expressing G. This means that the labeled input cells could be a mixed population connected with either the primary or secondary starter cells.

It would be immensely interesting if time-lapse imaging could be utilized to observe the appearance of such primary and secondary starter cells. Assuming there is a time difference

between the initial appearance of these two populations, it may be possible to differentiate the input cells wired to these populations based on a similar temporal difference in their initial appearance. This approach could provide valuable insights into the dynamics of rabies virus spread and the connectivity of neural circuits.

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Reviewer #2 (Public Review):

The study by Chen, Deng et al. aims to develop an efficient viral transneuronal tracing method that allows efficient retrograde tracing in the larval zebrafish. The authors utilize pseudotyped-rabies virus that can be targeted to specific cell types using the EnvA-TvA systems. Pseudotyped rabies virus has been used extensively in rodent models and, in recent years, has begun to be developed for use in adult zebrafish. However, compared to rodents, the efficiency of the spread in adult zebrafish is very low (~one upstream neuron labeled per starter cell). Additionally, there is limited evidence of retrograde tracing with pseudotyped rabies in the larval stage, which is the stage when most functional neural imaging studies are done in the field. In this study, the authors systematically optimized several parameters of rabies tracing, including different rabies virus strains, glycoprotein types, temperatures, expression construct designs, and elimination of glial labeling. The optimal configurations developed by the authors are up to 5-10 fold higher than more typically used configurations.

The results are solid and support the conclusions. However, the methods should be described in more detail to allow other zebrafish researchers to apply this method in their own work.

Additionally, some findings are presented anecdotally, i.e., without quantification or sufficient detail to allow close examinations. Lastly, there is concern that the reagents created by the authors will not be easily accessible to the zebrafish community.

(1) The titer used in each experiment was not stated. In the methods section, it is stated that aliquots are stored at 2×10^8 . Is it diluted for injection? Are all of the experiments in the manuscripts with the same titer?

1. The age for injection is quite broad (3-5 dpf in Fig 1 and 4-6 dpf in Fig 2). Given that viral spread efficiency is usually more robust in younger animals, describing the exact injection age for each experiment is critical.

(3) More details should be provided for the paired electrical stimulation-calcium imaging study. How many GC cells were tested? How many had corresponding PC cell responses? What is the response latency? For example, images of stimulated and recorded GCs and PCs should be shown.

(4) It is unclear how connectivity between specific PC and GC is determined for single neuron connectivity. In other images (Figure 4C), there are usually multiple starter cells and many GCs. It was not shown that the image resolution can establish clear axon-dendritic contacts between cell pairs.

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Reviewer #3 (Public Review):

Summary:

The authors establish reagents and define experimental parameters useful for defining neurons retrograde to a neuron of interest.

Strengths:

A clever approach, careful optimization, novel reagents, and convincing data together lead to convincing conclusions.

Weaknesses:

In the current version of the manuscript, the tracing results could be better centered with respect to past work, certain methods could be presented more clearly, and other approaches worth considering.

Appraisal/Discussion:

Trans-neuronal tracing in the larval zebrafish preparation has lagged behind rodent models, limiting "circuit-cracking" experiments. Previous work has demonstrated that pseudotyped rabies virus-mediated tracing could work, but published data suggested that there was considerable room for optimization. The authors take a major step forward here, identifying a number of key parameters to achieve success and establishing new transgenic reagents that incorporate modern intersectional approaches. As a proof of concept, the manuscript concludes with a rough characterization of inputs to cerebellar Purkinje cells. The work will be of considerable interest to neuroscientists who use the zebrafish model.

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Author response:

We are grateful to the reviewers for their insightful comments on our manuscript and are encouraged by their overall favorable assessments. For the eLife Version of Record, we will make the following revisions to address reviewers' comments and broaden the applicability of our technique in the zebrafish research community:

(1) We will elaborate on various facets with additional details:

a) Experimental conditions | We will specify the transgenic background, injected plasmids, larval stage, viral type, and viral titer clearly for each related experiment.

b) Experimental methods | We will depict in more details on how to inject the virus into a target area in larval zebrafish.

c) Data analysis | We will provide more detailed information on the paired electrical stimulation-calcium imaging study and on identifying connected Purkinje cells and granule cells during circuit reconstruction.

d) Discussion | We will elaborate on trans-synaptic specificity concerning glial cell labeling, toxicity related to viral dose and temperature, and the potential issue of secondary starters and multi-step circuit tracing.

(2) We will address the issue of glial cell labeling by adding more discussion and characterization, including potential mechanisms and implications, cell distribution, labeling progress, survival, and capability for viral transmission as starter cells.

(3) We will modify the text of the manuscript to clarify additional points raised by the reviewers.

(4) We will provide public repositories for accessing both the items and information on zebrafish lines, plasmids, viral vectors, and reconstructed data generated in this study.

In the end, we will submit full responses to the reviewer comments along with the revised version of the manuscript.

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